HOSPITAL PATIENT READMISSION ANALYSIS
Real-World Data Analytics Project Report

Project Title: Hospital Patient Readmission Analysis
Project Domain: Healthcare Data Analytics
Programming Language: Python
Development Environment: Google Colab and Visual Studio Code / Command Prompt
Dashboard Framework: Streamlit
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
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Academic Year: 2024-2028

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1. ABSTRACT

Hospital patient readmission is an important healthcare analytics problem. Understanding patterns in hospital encounters and subsequent readmissions can help identify areas for further investigation and support healthcare quality improvement.

This project, titled Hospital Patient Readmission Analysis, uses Python-based data analytics and machine-learning techniques to explore historical hospital encounter data. The project includes data analysis, visualisation, readmission pattern analysis, model development, and an interactive Streamlit dashboard.

The dashboard displays key indicators such as total hospital encounters, recorded readmissions, readmission within 30 days, and average hospital stay. It also provides filters for age groups and length of hospital stay.

The dashboard currently displays 55,226 hospital encounters, 46.7% with any recorded readmission, 11.4% readmission within 30 days, and an average hospital stay of 4.5 days. These figures will be verified against the dataset definitions and analysis code before being treated as final findings.

The project demonstrates how programming, data analysis, visualisation, and machine learning can be combined to investigate healthcare data. The results are intended for educational and analytical purposes and are not clinical recommendations.

2. INTRODUCTION

Healthcare organisations generate large amounts of data through patient admissions, hospital stays, diagnoses, treatments, and follow-up encounters. Analysing this information can reveal patterns that may otherwise be difficult to identify manually.

Patient readmission analysis focuses on understanding whether patients return to a hospital after an earlier encounter. Analysing readmission patterns by age group, hospital stay, and admission characteristics can help identify useful areas for further research.

This project develops an analytical workflow that uses Python for data processing and machine learning and Streamlit for interactive visualisation.

3. PROBLEM STATEMENT

Hospital readmission patterns can be difficult to understand when patient encounter records are examined individually. A systematic analytical approach is needed to summarise the data, identify patterns, compare relevant factors, and evaluate whether machine-learning models can predict a defined readmission outcome.

The main problem addressed by this project is how to analyse historical hospital encounter data and present meaningful, understandable results through an interactive dashboard.

4. OBJECTIVES

The objectives of the project are:

1. Load and inspect the hospital encounter dataset.
2. Identify and handle missing values, duplicate records, and data-quality issues.
3. Perform exploratory data analysis using Python.
4. Analyse readmission patterns across age groups, hospital stay durations, and admission types.
5. Prepare relevant features for machine-learning models.
6. Train and compare suitable classification algorithms.
7. Evaluate model performance using appropriate classification metrics.
8. Develop an interactive Streamlit dashboard.
9. Export analytical tables and model artefacts for further use.
10. Document findings, limitations, and possible future improvements.

5. TOOLS AND TECHNOLOGIES

- Python: Main programming language used for data analysis and modelling.

- Pandas: Loading, cleaning, transforming, and summarising tabular data.
- NumPy: Numerical operations and data processing.
- Matplotlib: Creating charts and visualisations.
- Seaborn: Statistical visualisation, where used.
- Scikit-learn: Machine-learning model training and evaluation.
- Streamlit: Building the interactive dashboard.
- Google Colab: Developing and executing notebook-based analysis.
- Visual Studio Code or a text editor: Editing the local application file, if used.
- Git and GitHub: Version control and project sharing, if used.

6. DATASET DESCRIPTION

The project uses a hospital-related dataset for encounter and readmission analysis.

Dataset file: [Confirm the exact dataset filename]

The dashboard currently reports 55,226 hospital encounters. This is the encounter count shown by the application and must be reconciled with the dataset and any filters applied.

The analysis includes fields or derived groupings related to age, hospital stay, admission type, readmission category, and other model features, subject to verification against the actual dataset columns.

The final report will document:

- Dataset source and licence, if available.
- Number of rows and columns in the original dataset.
- Meaning of the target variable.
- Definition of readmission within 30 days.
- Data types and missing-value counts.
- Any filtering or preprocessing performed.

7. METHODOLOGY

The project follows a structured data analytics workflow.

7.1 Data Loading

The dataset is loaded into Python using Pandas. The data is inspected to understand its dimensions, columns, data types, and sample records.

7.2 Data Cleaning

The data-cleaning stage checks for missing values, duplicate records, inconsistent data types, and invalid values. Any changes made to the data will be documented.

7.3 Exploratory Data Analysis

Exploratory data analysis is used to examine distributions and relationships in the data. Charts and summary tables help describe encounter patterns and readmission categories.

7.4 Feature Engineering

Relevant variables are prepared for analysis and modelling. Depending on the actual dataset, this may include age groups, hospital-stay features, categorical encoding, and the preparation of a clearly defined target variable.

7.5 Machine-Learning Development

Suitable classification models are trained to estimate the selected readmission outcome. The final report will name only the models confirmed in the notebook and document the train/test split and preprocessing method.

7.6 Model Evaluation

The model is evaluated using appropriate measures, such as precision, recall, F1-score, ROC-AUC, and a confusion matrix, where applicable. Evaluation results will be copied from the actual notebook outputs.

7.7 Dashboard Development

Streamlit is used to display key performance indicators, filters, and charts. The dashboard enables users to explore the historical data interactively.

8. DASHBOARD RESULTS

The current dashboard displays the following values:

Indicator	Displayed result
Hospital encounters	55,226
Any recorded readmission	46.7%
Readmission within 30 days	11.4%
Average hospital stay	4.5 days

These figures are transcribed from the running dashboard. They are preliminary until the calculation logic, dataset, filters, and outcome definitions are verified.

8.1 Dashboard Filters

The dashboard includes:

- Age-group selection.
- A hospital-stay duration slider.

These controls allow users to explore subsets of the displayed data.

8.2 Dashboard Visualisations

The visible dashboard includes charts for:

- Readmission categories.
- Readmission rates across age groups.

The application also exports analytical tables relating to age, hospital-stay duration, admission type, and feature importance, as indicated by the notebook output. The exact contents and calculations will be checked before final submission.

9. MACHINE-LEARNING RESULTS

Status: To be completed from the actual notebook outputs.

This section will include the names of the models trained, their evaluation metrics, a comparison table, a confusion matrix, and an interpretation of the results.

No model accuracy or performance score should be added until it has been obtained from the executed notebook.

10. KEY FINDINGS

The current dashboard provides a summary of hospital encounters and readmission indicators. It also allows exploration of patterns across age groups and hospital-stay durations.

The final findings will be written after verifying the underlying calculations, comparing relevant groups, and examining the model evaluation results. Any observed association will be described as an association rather than proof of causation.

11. LIMITATIONS

1. The findings depend on the quality and completeness of the available dataset.
2. Encounter counts may not correspond to unique patients if individuals have multiple encounters.

3. Readmission percentages depend on how readmission outcomes and observation windows are defined.
4. Predictive models may be affected by class imbalance, data leakage, or differences between training data and future data.
5. Historical associations do not establish causal relationships.
6. The dashboard is an educational analytics application and is not validated for clinical decision-making.

12. FUTURE SCOPE

Possible future improvements include:

- More extensive data-quality checks.
- Additional exploratory and statistical analyses.
- More rigorous model validation and comparison.
- Explainability analysis for model predictions.
- Improved dashboard filters and visualisations.
- A documented, reproducible data-processing pipeline.
- Automated updates when a suitable, authorised data source becomes available.

13. CONCLUSION

The Hospital Patient Readmission Analysis project demonstrates a workflow for exploring hospital encounter data using Python, visualisation libraries, machine learning, and Streamlit.

The interactive dashboard provides a convenient way to inspect readmission indicators and explore patterns across selected groups. The project also creates analytical outputs and a saved model artefact for further evaluation.

The final conclusions will depend on verification of the dataset, readmission definitions, and machine-learning evaluation results. The project is intended to demonstrate data analytics methods and does not replace clinical judgement.

14. REFERENCES

Add the actual references used in the project, such as:

- Documentation for Python and Pandas.
- NumPy documentation.
- Matplotlib documentation.
- Scikit-learn documentation.
- Streamlit documentation.
- The original dataset source and its documentation.